

Vaansh Sanghrajka

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Education

Stevens Institute of Technology

Expected - May 2027

MS in Computer Science

University of Mumbai, India

Sept 2022 – May 2025

Bachelor of Technology (B.Tech) - Information Technology

Honours in Artificial Intelligence and Machine Learning

Experience

AI & ML Engineer Intern | Aarvi Technology, India

July 2024 – Nov 2024

- Applied statistical analysis and machine learning techniques to evaluate model performance and improve prediction accuracy by 25%.
- Performed data preprocessing, feature engineering, and exploratory data analysis using Pandas and NumPy to support reliable model training.
- Conducted quantitative model evaluation using validation techniques and performance metrics to ensure correctness and consistency.

Programming Teaching Assistant | Programmatix Institution

Jan 2024 – April 2024

- Assisted students with problem solving in algorithms, discrete mathematics, and statistical reasoning concepts.
- Evaluated student assignments and provided structured feedback on correctness, logic, and implementation quality.
- Helped students debug programs and understand mathematical foundations behind algorithm performance.

Projects

AI Code Analysis Assistant (Personal Research Project)

Jan 2026 – Present

- Developing an LLM-based assistant to analyze software code structure and detect logical inefficiencies using structured evaluation methods.
- Applied analytical reasoning to identify redundant logic patterns and improve maintainability through systematic code review approaches.
- Designed evaluation workflows to measure code quality improvements and consistency of generated solutions.

TradeMentor – News Driven Real Time Trading Simulator

Sept 2025 – Dec 2025

- Built a real-time trading simulator analyzing market behavior using data-driven decision logic and event-driven architecture.
- Applied statistical reasoning to simulate price movements based on sentiment signals and market events.
- Designed data structures to track portfolio performance and evaluate trading outcomes using quantitative metrics.

Agentic LLM System for Code Generation and Reasoning

Sept 2024 – May 2025

- Developed a transformer-based system for structured problem solving and multi-step reasoning tasks.
- Designed evaluation experiments to analyze reasoning consistency, correctness, and response reliability.
- Applied quantitative evaluation techniques to measure model performance improvements.
- Optimized inference workflows improving response latency by 35%.

Technical Skills

Programming: Python, C, C++, Java, SQL, Object-Oriented Programming, Data Structures & Algorithms

Statistics & Mathematics: Probability Theory, Statistical Analysis, Hypothesis Testing, Regression, Descriptive Statistics, Variance Analysis, Data Interpretation

Data Analysis: Pandas, NumPy, Data Cleaning, Data Validation, Exploratory Data Analysis (EDA), Model Evaluation Metrics

Academic Support Skills: Assignment Grading, Solution Verification, Mathematical Problem Solving, Attention to Detail, Student Mentoring

AI & Machine Learning: Machine Learning Fundamentals, Model Evaluation, Feature Engineering, Scikit-learn, TensorFlow, Prompt Engineering

Software Engineering: Debugging, Code Optimization, Refactoring Concepts, Software Design Principles, Git

Tools: Microsoft Excel (Pivot Tables, VLOOKUP, Data Analysis), Tableau, Power BI, Jupyter Notebook, GitHub

Cloud: AWS (EC2, S3)